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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- The number of solutions of $5^{1+|\sin x|+|\sin x|^2+\dots} = 25$
For $x \in (-\pi, \pi)$ is
(a) 2 (b) 0
(c) 4 (d) infinite
- Consider the function $f(x) = \begin{cases} -x^3 + 3x^2 + 1, & \text{if } x \leq 2 \\ \cos(x) & \text{if } 2 < x \leq 4 \\ e^{-x} & \text{if } x > 4 \end{cases}$
Which of the following statement about $f(x)$ is true
(a) $f(x)$ has a local maximum at $x = 1$, which is also the global maximum
(b) $f(x)$ has a local maximum at $x = 2$, which is not the global maximum.
(c) $f(x)$ has a local maximum at $x = \pi$, but it is not the global maximum.
(d) $f(x)$ has a global maximum at $x = 0$.
- The two parabolas $y^2 = 4a(x+c)$ and $y^2 = 4bx$, $a > b > 0$ cannot have a common normal unless
(a) $c > 2(a+b)$ (b) $c > 2(a-b)$
(c) $c < 2(a-b)$ (d) $c < 2/a-c$
- Let Z be the set of all integers, and consider the sets $X = \{(x,y) : x^2 + 2y^2 = 3, x, y \in Z\}$ and $Y = \{(x,y) : x > y, x, y \in Z\}$. Then the number of elements in $X \cap Y$ is
(a) 2 (b) 1
(c) 3 (d) 4
- Region R is defined as region in first quadrant satisfying the condition $x^2 + y^2 < 4$. Given that a point $P = (r, s)$ lies in R , what is the probability that $r > s$?
(a) 1 (b) 0
(c) $1/2$ (d) $1/3$
- Let $f : R \rightarrow R$ be a function such that $f(0) = \frac{1}{\pi}$ and $f(x) = \frac{x}{e^{\pi x} - 1}$ for $x \neq 0$
(a) $f(x)$ is not continuous at $x = 0$
(b) $f(x)$ is continuous but not differentiable at $x = 0$
(c) $f(x)$ is differentiable at $x = 0$ and $f'(0) = -\frac{\pi}{2}$
(d) None of these
- Consider the function $f(x) = x^{2/3}(6-x)^{1/3}$. Which of the following statement is FALSE?
(a) f is increasing in the interval $(0, 4)$
(b) f is decreasing in the interval $(6, \infty)$
(c) f has a point of inflection at $x = 0$
(d) f has a point of inflection at $x = 6$
- The vector $\vec{A} = (2x+1)\hat{i} + (x^2-6y)\hat{j} + (xy^2+3z)\hat{k}$ is a
(a) sink field (b) solenoidal field
(c) source field (d) None of these
- For an invertible matrix A , which of the following is not always true:
(a) $|\text{adj}(A)| \neq 0$ (b) $|A| \neq 0$
(c) $|AA^{-1}| = 1$ (d) $|A(\text{adj}(A))| \neq 1$

- The value of the limit $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x + 4^x}{4} \right)^{1/x}$ is
(a) 1 (b) $3!^{1/3!}$
(c) $3!^{1/4}$ (d) $4!^{1/4}$

- At how many points the following curves intersect
 $\frac{y^2}{9} - \frac{x^2}{16} = 1$ and $\frac{x^2}{4} + \frac{(y-4)^2}{16} = 1$
(a) 0 (b) 1
(c) 2 (d) 4

- The value of $\tan\left(\frac{\pi}{4} + \theta\right) \tan\left(\frac{3\pi}{4} + \theta\right)$ is
(a) -2 (b) 2
(c) 1 (d) -1

- The system of equations $x + 2y + 2z = 5$, $x + 2y + 3z = 6$, $x + 2y + \lambda z = \mu$ has infinitely many solution if
(a) $\lambda \neq 2$ (b) $\lambda \neq 2, \mu \neq 5$
(c) $\lambda = 2, \mu = 5$ (d) $\mu \neq 5$

- The value of $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{1 - \cos x}$ is equal to
(a) 2 (b) 1
(c) 0 (d) -1

- If the line $ax + ay + 1 = 0$, for some real number a , is normal to the curve $xy = 1$, then
(a) $a < 0$ (b) $0 < a < 1$
(c) $a > 0$ (d) $-1 < a < 1$

- A coin is thrown 8 number of times. What is the probability of getting a head in an odd number of throw?
(a) $3/4$ (b) $1/4$
(c) $1/2$ (d) $1/8$

- A critical orthopedic surgery is performed on 3 patients. The probability of recovering a patient is 0.6. Then the probability that after surgery, exactly two of them will recover is
(a) 0.123 (b) 0.432
(c) 0.321 (d) 0.234

- The value of $f(1)$ for $f\left(\frac{1-x}{1+x}\right) = x + 2$ is
(a) 1 (b) 2
(c) 3 (d) 4

- Let C denote the set of all tuples (x,y) which satisfy $x^2 - 2y$ where x and y are natural numbers. What is the cardinality of C ?
(a) 0 (b) 1 (c) 2 (d) 3

- If $f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$, where $[.]$ stands for the greatest integer function, then $f(\pi/2) =$
(a) -1 (b) 0 (c) 1 (d) 2

- If $\sin x = \sin y$ and $\cos x = \cos y$, then the value of $x - y$ is
(a) $\pi/4$ (b) $n\pi/2$
(c) $n\pi$ (d) $2n\pi$



22. Lines $L_1, L_2, \dots, L_{10}$ are distinct among which the lines $L_2, L_4, L_6, L_8, L_{10}$ are parallel to each other and the lines L_1, L_3, L_5, L_7, L_9 pass through a given point C. The number of point of intersection of pairs of lines from the complete set $L_1, L_2, L_3, \dots, L_{10}$ is:

- (a) 24 (b) 25
(c) 26 (d) 27

23. If the perpendicular bisector of the line segment joining $p(1,4)$ and $q(k,3)$ has y-intercept -4, then the possible value of k are

- (a) -4 and 4 (b) -2 and 2
(c) -3 and 3 (d) -1 and 1

24. Out of a group of 50 students taking examinations in Mathematics, Physics, and Chemistry, 37 students passed Mathematics, 24 passed Physics, and 43 passed Chemistry. Additionally, no more than 19 students passed both Mathematics and Physics, no more than 29 passed both Mathematics and Chemistry, and no more than 20 passed both Physics and Chemistry. What is the maximum number of students who could have passed all three examinations?

- (a) 14 (b) 12
(c) 10 (d) 9

25. If one AM (Arithmetic mean) 'a' and two GM's (Geometric means) p and q be inserted between any two positive numbers, the value of $p^3 + q^3$ is

- (a) $2 a p q$ (b) $p q/a$
(c) $2 p q/a$ (d) $p + q + a$

26. Find the given numbers below, the smallest number which will be divided by 9, 10, 15 and 20, leaves the remainders 4, 5, 10 and 15 respectively

- (a) 265 (b) 535
(c) 355 (d) 85

27. It is given that the mean, median and mode of a data set is 1, $3x$ and $9x$ respectively. The possible values of the mode is

- (a) 1,4 (b) 1,9
(c) 3, 9 (d) 9, 8

28. A man starts at the origin O and walks a distance of 3 units in the north-east direction and then walks a distance of 4 units in the north-west direction to reach the point P, then $\overline{OP}$ is equal to

- (a) $\frac{1}{\sqrt{2}}(-\hat{i} + \hat{j})$ (b) $\frac{1}{2}(\hat{i} + \hat{j})$
(c) $\frac{1}{\sqrt{2}}(\hat{i} - 7\hat{j})$ (d) $\frac{1}{\sqrt{2}}(-\hat{i} + 7\hat{j})$

29. The points $(1, 1/2)$ and $(3, -1/2)$ are

- (a) In between the lines $2x + 3y = 6$ and $2x + 3y = -6$
(b) on the same side of the line $2x + 3y = 6$
(c) on the same side of the line $2x + 3y = -6$
(d) on the opposite side of the line $2x + 3y = -6$

30. If $(4, 3)$ and $(12, 5)$ are the two foci of an ellipse passing through the origin, then the eccentricity of the ellipse is

- (a) $\frac{\sqrt{13}}{9}$ (b) $\frac{\sqrt{13}}{18}$
(c) $\frac{\sqrt{17}}{18}$ (d) $\frac{\sqrt{17}}{9}$

31. The value of series

$$\frac{2}{3!} + \frac{4}{5!} + \frac{6}{7!} + \dots$$

- (a) $2e^{-2}$ (b) e^{-2}
(c) e^{-1} (d) $2e^{-1}$

32. There are 9 bottle labelled 1,2,3.....9 and 9 boxes labelled 1,2,3,.....9. The number of ways one can put these bottles in the boxes so that each box gets one bottle and exactly 5 bottles go in their corresponding numbered boxes is

- (a) $9 \times {}^9C_5$ (b) $5 \times {}^9C_5$
(c) $25 \times {}^9C_5$ (d) $4 \times {}^9C_5$

33. A committee of 5 is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all is

- (a) $5/9$ (b) $1/2$
(c) $2/3$ (d) $4/9$

34. If $x = 1 + \sqrt[5]{2} + \sqrt[5]{4} + \sqrt[5]{8} + \sqrt[5]{16} + \sqrt[5]{32}$, then $\left(1 + \frac{1}{x}\right)^{24} =$

- (a) 1 (b) 4
(c) 16 (d) 24

35. For what values of λ does the equation $6x^2 - xy + \lambda y^2 = 0$ represents two perpendicular lines and two lines inclined at an angle of $\pi/4$.

- (a) -6 and -2 (b) 6 and 1
(c) -6 and -35 (d) -6 and 1

36. If $|F| = 40$ N (Newtons), $|D| = 3$ m, and $\theta = 60^\circ$, then the work done by F acting from P to Q is

- (a) $60\sqrt{3}$ J (b) 120J
(c) $60\sqrt{2}$ J (d) 60J

37. If three distinct numbers are chosen randomly from the first 100 natural numbers, then the probability that all three of them are divisible by both 2 and 3 is

- (a) $4/35$ (b) $4/33$
(c) $4/1155$ (d) $4/25$

38. The value of m for which volume of the parallelepiped is 4 cubic units whose three edges are represented by $a = \hat{i} + \hat{j} + \hat{k}$, $b = \hat{i} - \hat{j} + \hat{k}$, $c = \hat{i} + 2\hat{j} - \hat{k}$ is

- (a) 0 (b) -2 (c) -1 (d) 1

39. Let $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Then which of the following is true

- (a) $f(x)$ is not continuous at $x = 0$
(b) $f(x)$ is not differentiable at $x = 0$
(c) $f'(x)$ is not continuous at $x = 0$
(d) $f'(x)$ is continuous at $x = 0$

40. Which of the following is TRUE?

- (a) If f is continuous on $[a, b]$, then $\int_a^b x f(x) dx = x \int_a^b f(x) dx$
(b) $\int_0^3 e^{x^2} dx = \int_0^5 e^{x^2} dx + \int_5^3 e^{x^2} dx$
(c) If f is continuous on $[a, b]$, then $\frac{d}{dx} \left(\int_a^b f(x) dx \right) = f(x)$
(d) Both (a) and (b)

41. Given a set A with median $m_1 = 2$ and set B with median $m_2 = 4$ What can we say about the median of the combined set?

- (a) at most 1 (b) at most 2
(c) at least 1 (d) at least 2



42. The value of $\sum_{r=1}^n \frac{1}{2^n} \frac{nPr}{r!}$ is:

- (a) 2^n (b) $1-2^{-n}$
(c) 2^n-1 (d) $2^{2n}-1$

43. The number of one-one function

$f: \{1,2,3\} \rightarrow \{a, b, c, d, e\}$ is

- (a) 125 (b) 60
(c) 243 (d) None of the above

44. Find the cardinality of the set C which is defined as

$$C = \{x | \sin 4x = \frac{1}{2} \text{ for } x \in (-9\pi, 3\pi)\}$$

- (a) 24 (b) 48
(c) 36 (d) 12

45. How much work does it take to slide a crate for a distance of 25 m along a loading dock by pulling on it with a 180 N force where the dock is at an angle of 45° from the horizontal?

- (a) 3.18198×10^3 J
(b) 3.18198×10^2 J
(c) 3.4341×10^3 J
(d) 3.4341×10^4 J

46. If for non-zero x, $cf(x) + df(1/x) = |\log|x|| + 3$, where $c \neq d$, then $\int_1^e f(x)dx =$

- (a) $\frac{(c-d)(2e-1)}{c^2-d^2}$ (b) $\frac{(c-d)(3e-2)}{c^2-d^2}$
(c) $\frac{(c-d)(3e+2)}{c^2-d^2}$ (d) $\frac{(c-d)(2e+1)}{c^2-d^2}$

47. The equation $3x^2 + 10xy + 11y^2 + 14x + 12y + 5 = 0$ represent,

- (a) a circle (b) an ellipse
(c) a hyperbola (d) a parabola

48. Let A and B be two events defined on a sample space Ω . Suppose A^c the complement of A relative to the sample space Ω . Then the probability $P((A \cap B^c) \cup (A^c \cap B))$ equal

- (a) $P(A) + P(B) + P(A \cap B)$
(b) $P(A) + P(B) - P(A \cap B)$
(c) $P(A) + P(B) + 2P(A \cap B)$
(d) $P(A) + P(B) - 2P(A \cap B)$

49. The number of distinct real values of $\lambda^2 \hat{i} + \hat{j} + \hat{k}$, $\hat{i} + \lambda^2 \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + \lambda^2 \hat{k}$ are coplanar is

- (a) 1 (b) 2
(c) 3 (d) 6

50. A speaks truth in 40% and B in 50% of the cases. The probability that they contradict each other while narrating some incident is

- (a) $2/3$ (b) $1/2$
(c) $1/3$ (d) 0

REASONING

51. Which pairs of bits can be joined together to form two words that have opposite meanings?

ER	UC	DE	EN	EX	EA	AN	SI	RE	GO
T	E	S	D	P	R	D	P	D	S
1	2	3	4	5	6	7	8	9	10

- (a) (9,2) (5,7) (b) (1,3) (8,10)
(c) (1,5) (10,8) (d) (4,2) (7,8)

52. Rajesh will not go to the concert if Rakesh goes. Rakesh will go to the concert if his dog barks three times. Based only on the information above, which of the following must be true?

- (a) Rakesh will not go to the concert unless Rajesh goes.
(b) If Rakesh's dog does not bark three times, then Rakesh will not go to the concert
(c) If Rakesh's dog barks three times, then Rajesh will not go to the concert
(d) If Rajesh doesn't go to the concert, then Rakesh will go

53. Which one of the following is the odd one from the given alternative?

- (a) years of experience (b) Salary
(c) Highest education (d) Age

54. Looking at the portrait of a man, Lucky (male) said, "This person is the only child of my paternal grandmother's daughter." Whose portrait was Lucky looking at?

- (a) His brother (b) His cousin
(c) His uncle (d) Himself

55. Study the following diagram and answer the following question



By which letter, the married teachers who do not live in joint family are represented?

- (a) Q (b) P (c) R (d) S

56. Study the following information carefully and answer the given question:

Eight friends A, B, C, D, E, F, G and H are sitting on a round table facing the centre. A sits second to the left of D, who sits third to the left of E. C sits third to the right of G, who is not an immediate neighbour of E. H sits opposite to the E. B is between A and C.

- (a) E (b) D (c) G (d) F

57. From the given options, find the pair which is like the given pair 8:4

- (a) 27: 9 (b) 216: 32
(c) 72: 24 (d) 45: 5

58. Select the pair of words, which are related in the same way as the capitalized words are related to each other.

DATA : GRAPH

- (a) mother : Father (b) Plant : Leaf
(c) Milk : Butter (d) Water : Glass

59. A cat climbs a 21- meter pole. In the first minute it climbs 3 meter and in the second minute it descends one meter. In how minutes the cat would reach the top of the pole?

- (a) 18 min (b) 21 min
(c) 19 min (d) 20 min

60. Arrange the words given below in a meaningful sequence

- (1) Software (2) Code (3) Data (4) Analysis (5) Report
(a) 5,4,3,1,2 (b) 2,1,5,3,4
(c) 3,1,2,5,4 (d) 3,1,2,4,5



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61. The question contains six statements followed by four sets of combinations of three. Choose the set in which the combinations are most logically related.
- A. Some buildings are not sky scrapers.
B. Some sky scrapers are not buildings.
C. No structure is a skyscraper.
D. All skyscrapers are structures.
E. Some skyscrapers are buildings
F. Some structures are not buildings.
- (a) ACF (b) ACE (c) BDF (d) FDA
62. Select the one which is different from the other three.
- (a) HEM (b) JGP (c) OLT (d) NKS
63. Two cars, Car A and Car B, are travelling on a highway. Car A starts from point X and travels at a constant speed of 60 km/h, while Car B starts from the same point X but travels at a constant speed of 80 km/hr. If both cars travel for 1.5 hours, what is the difference in distance covered by Car B compared to Car A?
- (a) 20 km (b) 25 km (c) 35 km (d) 30 km
64. After allowing 20% cash discount, a trader still earns a profit of 11.11%. How much above the cost price, the trader marks his goods?
- (a) 38.88% (b) 40% (c) 30.33% (d) 28%
65. In the following question, three statements and three conclusions are given.
- Statements:
- All students are intelligent.
 - No intelligent person is lazy.
 - Some lazy people are poor.
- Conclusions:
- No student is lazy.
 - Some poor people are not intelligent.
 - All poor people are lazy.
- Find out the most appropriate conclusion(s) from the following options.
- (a) Only conclusion 2 follows.
(b) Only conclusions 2 and 3 follow
(c) Only conclusion 1 follows
(d) Only conclusions 1 and 2 follows
66. In which year was Arjun born?
- Arjun at present is 25 years younger to his mother.
Arjun's brother, who was born in 1964, is 35 years younger to his mother.
- (a) 1974 (b) 1964 (c) 1944 (d) 1954
67. You are on an island with two tribes. One tribe always tells the truth, and the other tribe always lies. You meet three individuals from the island A, B and C. Each individual belongs to one of the tribes. You ask each of them the same question "Is B a truth-teller?"
- A says, "Yes B is a truth-teller."
B says, "No, I am not a truth-teller"
C. Says, "B is a liar"
- Given that each individual is either a truth-teller or a liar, who is telling the truth?
- (a) A only (b) Both B and C
(c) C only (d) B only

68. In a recent survey of 500 employees in a company, it was found that 60% of the employees prefer coffee over tea, 25% prefer tea over coffee, and the remaining 15% have no preference. If 20% of the employees who prefer coffee are also tea drinkers, how many employees prefer only tea?
- (a) 75 (b) 50 (c) 65 (d) 55
69. In certain language, HEART is written as 2018010508, and LUNGS is written as 1907142112. If Brain is written in that language, what will be the last number?
- (a) 2 (b) 5 (c) 4 (d) 9
70. The mean of consecutive positive integers from 2 to n is
- (a) $\frac{n+2}{2}$ (b) $\frac{n(n+1)}{2}$ (c) $\frac{n+1}{2}$ (d) $\frac{n-1}{2}$
71. In a tournament, how many teams participated. All teams in the tournament have 5 to 15 players. If a team has more than 10 players, then they have reversible t-shirts?
- (a) Teams with 8 players do not have reversible t-shirts
(b) Teams that have 12 players do not have reversible t-shirts.
(c) Only people on teams can have reversible t-shirts.
(d) Teams that have 13 players have reversible t-shirts.
72. If by rearranging the letters of the word NABMODINT, a name of a game is formed. What would be the first and last letter of the mirror image of the name of the game?
- (a) T, B (b) B, N (c) B, T (d) N, B
73. What is the value of $x^2 + y^2 = ?$
- Statement I: $xy = 5$
Statement II: $x + y = 10$
- A. Choose this option if the question can be answered by using one of the statement alone, but cannot be answered using the other statement.
B. Choose this option if the question can be answered by using the statement together, but cannot be answered using either statement alone.
C. Choose this option if the question can be answered by using either statement alone.
D. Choose this option if the question cannot be answered even by using both the statement together.
- (a) B (b) C (c) D (d) A
74. Four friends, Aditi, Bharat, Chandan, and Deepika, went to a restaurant for dinner. Each of them ordered a different dish from the menu: pizza, pasta, burger, and salad. Additionally, each friend ordered a different drink: cola, lemonade, orange juice, and water. Based on the following clues, determine the combination of friend, dish, and drink:
- Aditi didn't order pizza or cola.
 - Bharat ordered salad but not lemonade.
 - Chandan ordered pasta.
 - Deepika didn't order burger or orange juice.
 - Aditi ordered orange juice.
- Who ordered the burger, and what drink did they order?
- (a) Deepika, cola (b) Bharat, water
(c) Aditi, orange juice (d) Chandan, lemonade





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75. This question contains six statemnt followed by four sets of combinations of three. Choose the set in which the combinations are most logically related.
- A. All falcons fly high.
B. All falcons are blind.
C. All falcons are birds.
D. All birds are yellow.
E. All birds are thirsty.
F. All falcons are yellow.
- (a) DEF (b) CDF
(c) ABC (d) BCA
76. In the half yearly exam only 60% of the students were passed. Out of these (passed in halfyearly) only 70% students are passed in annual exam, out of remaining students (who fail in half-yearly exam) 80% passed in annual exam. What percent of the students passed the annual exam?
- (a) 76% (b) 74%
(c) 72% (d) 65%
77. Odometer is to mileage as compass is to
- (a) Needle (b) Hiking
(c) Speed (d) Direction
78. Which of the following words will appear last in the dictionary.
- (a) complicit (b) compline
(c) compliment (d) complete
79. Aryan bought 100 shares of a company at Rs. 50 per share. He paid a brokerage fee of 2% on the purchase. Later, he sold all the shares at Rs. 55 per share and paid a brokerage fee of 2% on the sale. What is Aryan's net profit percentage on his investment?
- (a) 5.5% (b) 6%
(c) 5.69% (d) 6.1%
80. Ramu visits Delhi on every 15 days and Samu goes to Delhi every 20 days. They met at Delhi 5 days back. After how many days, from today, they will meet at Delhi next time?
- (a) 35 (b) 55
(c) 60 (d) 65
81. If 30th September, 1991 was a Wednesday, then what was the day on 14th March 1992?
- (a) Wednesday (b) Monday
(c) Saturday (d) Sunday
82. In a reality show, two judges independently provided marks base do the performance of the participants. If the marks provided by the second judge are given by $Y = 10.5 + 2x$, where X is the marks provided by the first judge. If the variance of the marks provided by the second judge is 100, then the variance of the marks provided by the first judge is:
- (a) 49.5 (b) 25
(c) 50 (d) 99

83. At what time between 2 pm and 3 pm, will the hour and minute hands of a clock point in opposite directions (diametrically opposite)?
- (a) 2:45 pm (b) 2:44 pm
(c) $2:43 \frac{9}{11}$ pm (d) $2:43 \frac{7}{11}$ pm

Comprehension: 84 to 86:

Directions: A B C D E F, and G are travelling in three different vehicles. There are at least two passengers in each vehicle- Swift, Creta, Nexon, and only one of them is a male. There are two engineers, two doctors and three teachers among them.

- C is a lady doctor and she does not travel with the pair of sisters A and F.
- B, a male engineer, travels with only G, a teacher in a Swift.
- D is a male doctor.
- Two persons belonging to the same profession do not travel in the same vehicle.
- A is not an engineer and travels in a Creta.
- The pair of sisters A and F travel in the same vehicle.

84. What is F's profession?
- (a) Doctor (b) Data inadequate
(c) Teacher (d) Engineer
85. In which vehicle does C travel?
- (a) Creta (b) Swift
(c) Nexon (d) Data inadequate
86. Which of the following represents the three teachers?
- (a) Data inadequate (b) GEA
(c) GEF (d) GBF

COMPREHENSION:

Direction : A, B, C, D and E are five different integer. When written in the ascending order of values, the difference between any two adjacent integers is 8. D is the greatest and A the least. B is greater than E but less than C. The sum of the integer is equal to E.

87. The value of A is the least
- (a) -18 (b) -15
(c) none of these (d) -17
88. the sum of A and B is
- (a) -30 (b) -15
(c) None of these (d) -20
89. The greatest number has the value
- (a) 14 (b) 12
(c) 15 (d) 17
90. The sum of the integers is
- (a) None of these (b) -10
(c) -8 (d) -6

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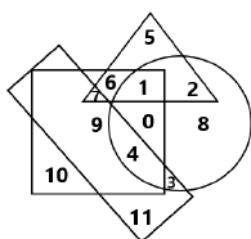
COMPUTER

91. The quotient, if the binary numebr 11010111 is divided by 101, is__
 (a) 101101 (b) 101010
 (c) 111001 (d) 101011
92. Any given truth table can be represented by
 (a) a Karnaugh map
 (b) a product of sum Boolean expression
 (c) a sum of product Boolean expression
 (d) all of the options
93. Which of the following interfaces perform the transfer of data between the memory and the I/O peripheral without involving the CPU?
 (a) DMA (b) Serial Interface
 (c) branch interface (d) DDA
94. Cache memory functions as an intermediary between
 (a) CPU and Hard Disk
 (b) None of these
 (c) CPU and RAM
 (d) RAM and ROM
95. A CPU generates 32 bits virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128-page table entires and is 4- way set associate. The minimum size of the TLB tag is
 (a) 11 bits (b) 20 bits
 (c) 13 bit (d) 15 bit
96. The expression $P+QR$ is the reduced form of ____
 (a) $(P + R)Q$ (b) $(P+Q) (P+R)$
 (c) $PQ + QR$ (d) $(P+Q) R$
97. The primary purpose of cache memory in a computer system is
 (a) to temporarily store frequently accessed data and instruction for faster access by the CPU.
 (b) To permanentaly store data and programs
 (c) to provide additional storage space when the main memory is full.
 (d) to manage input and output operations between the CPU and peripherals.
98. Consider the program below which uses six temporary variables a, b, c, d, e, and f.
 $a = 10$
 $b = 20$
 $c = 30$
 $d = a + c$
 $e = b + d$
 $f = c + e$
 $b = c + e$
 $e = b + f$
 $d = 5 + e$
 return d + f
- Assuming that all the above operations take their operands from registers, the minimum number of

- registers needed to execute this program without spilling is
 (a) 6 (b) 4
 (c) 5 (d) 3
99. Which of the following is the smallest unit of data in a computer ?
 (a) Bit (b) Byte
 (c) KB (d) Nibble
100. Consider the following 4- bit binary numbers represented in the 2's complement form : 1101 and 0100
 What would be the result when we add them?
 (a) 0001 and no overflow
 (b) 1001 and an overflow
 (c) 0001 and an overflow
 (d) 1001 and no overflow
101. A computer system has 16-bit wide address/ data bus that uses RAM chips of 4K x 8-bit capacity. The numebr of RAM chips are needed to provide a memory capacity of 64 Kbytes memory is
 (a) 32 (b) 16
 (c) 64 (d) 8
102. Given that numbers A and B are two 8 bit 2's complement numbers with $A = 11111111$, $B = 11111111$. Then sum $A + B$ is ____
 (a) 00000010 (b) 11111100
 (c) 11111110 (d) 00000000
103. The Boolean expression for the following truth table is
- | x | y | z | f |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |
- (a) $F = x'yz' + xy'z + x'y'z'$
 (b) $F = x'y'z' + xy'z + xyz'$
 (c) $F = x'yz' + xy'z + xyz$
 (d) None of these
104. Consider an arbitrary number system with independent digits as 0,1 and A. If we generate first few numbers in sequence as 00,01, 0A, 10, 11, 1A and if this process is continued to generate the numbers, then the position of 10A is ____
 (a) 15 (b) 10 (c) 12 (d) 9
105. Which of the following components is not a part of an instruction formation in CPU processing?
 (a) Opcode (b) Register file
 (c) Source operand (d) Destination operand
106. The range of the exponent E in the IEEE754 double precision (Binary 64) format is ____
 (a) $-1023 \leq E \leq 1022$
 (b) $-1022 \leq E \leq 1022$
 (c) $-1022 \leq E \leq 1023$
 (d) $-1023 \leq E \leq 1023$



107. Which of the following components is used to establish a communication link between a CPU and the peripheral devices to transfer data?
- index register
 - memory address register
 - memory data register
 - instruction register
108. In the figure, the circle stands for employed, the square stands for a social worker, the triangle stands for illiterate, and the rectangle stands for truthful. Study the figure with its regions and find the number of neither truthful nor illiterate people among the employed only.



- 4
 - 8
 - 11
 - 1
109. Which of the following do not affect CPU performance?
- number of cores
 - Cache size
 - Amount of RAM
 - Clock speed
110. Let the given number 11001, 1001 and 111001 be correspond to the 2's complement representation. Then with which one of the following decimal number, the given numbers match
- 6, -6 and -6, respectively
 - 7, -7 and -7 respectively
 - 25, -9 and -57 respectively
 - 25, 9 and 57, respectively

ENGLISH

111. Select the correct meaning of 'Peruse':
- Examine
 - Pursue
 - Continue
 - Rescue
112. Identify the word that means the same as "ostentatious".
- Lavish
 - Simple
 - Modest
 - Unassuming

113. Write the antonym for 'Inscrutable'
- Mysterious
 - obscure
 - comprehensible
 - Opaque
114. Choose the correct combination of preposition to complete the sentence:
"The cat jumped ___the table___ the chair."
- into, beside
 - onto, towards
 - on, from
 - off, in
115. Which sentence demonstrates correct preposition usage?
- Prefer coffee over tea
 - He is interested on learning new languages
 - She arrived to the party at 8 pm.
 - They were surprised of the sudden announcement
116. Select the appropriate synonym for 'coercive'
- Forceful
 - corrective
 - Merciful
 - Gentle
117. Choose the best option that indicates the change of voice for the sentence given below:
- Had you invited Alice?
 - Was Alice invited you?
 - Did you invited by Alice?
 - Were you invited by Alice?
118. The company's _____ growth in revenue surprised analysts.
- erratic
 - gradual
 - Exponential
 - Stagnant
119. What does the idiom "jump on the bandwagon" mean?
- To start a business
 - To repair a vehicle
 - To criticize something unfairly
 - To join a popular trend or activity
120. Which of the following is an essential element of a technical report?
- Creative storytelling
 - Anecdotes and personal opinions
 - Anecdotes and personal opinions
 - Emotional appeals





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ANSWER KEY

1	c	2	b	3	b	4	b	5	c
6	b	7	b	8	a	9	d	10	d
11	c	12	d	13	c	14	c	15	a
16	c	17	b	18	b	19	c	20	a
21	d	22	c	23	a	24	a	25	a
26	c	27	a	28	d	29	a	30	d
31	c	32	a	33	d	34	c	35	c
36	d	37	c	38	d	39	c	40	b
41	d	42	b	43	b	44	b	45	d
46	b	47	b	48	d	49	b	50	b
51	a	52	c	53	c	54	b	55	c
56	d	57	a	58	c	59	c	60	d
61	c	62	b	63	d	64	a	65	d
66	d	67	c	68	c	69	a	70	a
71	d	72	d	73	b	74	c	75	b
76	b	77	d	78	b	79	c	80	b
81	b	82	b	83	a	84	d	85	c
86	b	87	a	88	d	89	a	90	b
91	d	92	d	93	a	94	c	95	d
96	b	97	a	98	d	99	a	100	a
101	b	102	c	103	c	104	d	105	b
106	c	107	c	108	b	109	c	110	b
111	a	112	a	113	c	114	b	115	a
116	a	117	d	118	c	119	d	120	b

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SOLUTION

MATHS

1. The number of solution of $5^1 + |\sin x| + |\sin x|^2 + \dots = 25$

For $x \in (-\pi, \pi)$

$$5^1 + |\sin x| + |\sin x|^2 + \dots = 25 \dots$$

$$= 5^2$$

$$1 + |\sin x| + |\sin x|^2 + \dots = 2$$

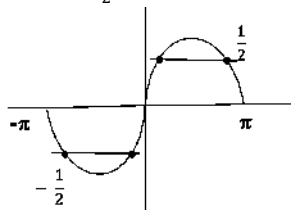
$$|\sin x| + |\sin x|^2 + \dots = 1$$

$$\frac{|\sin x|}{1 - |\sin x|} = 1$$

$$|\sin x| = 1 - |\sin x|$$

$$2|\sin x| = 1$$

$$\sin x = \pm \frac{1}{2}$$



Total 4 Solution.

$$2. f(x) = \begin{cases} -x^3 + 3x^2 + 1 & x \leq 2 \\ \cos x & 2 < x < 4 \\ e^{-x} & x > 4 \end{cases}$$

$$f'(x) = \begin{cases} -3x^2 + 6x & x < 2 \\ -\sin x & 2 < x < 4 \\ -e^{-x} & x > 4 \end{cases}$$

$f'(x) = 0$ is local maximum

$$-3x^2 + 6x = 0$$

$$x = 0$$

$$f''(x) = -6x$$

at $x = 2$ is local maximum

$$-\sin x = 0$$

$$x = \pi$$

$$= -\cos \pi = +1 \rightarrow \min$$

so $f(x)$ has local maximum at $x = 2$ which is not global max.

3. (b)

$$y^2 = 40(x+c) \dots \dots \dots$$

$$y^2 = 4b x \dots \dots \dots$$

The eq. of normal for 1

$$y = m(x+c) - (am^3 + 2am)$$

for eq. 2 normal

$$y = mx - (bm^3 + 2bm)$$

If they are common normal then

$$mx - (bm^3 + 2bm) = m(n+c) - (am^3 + 2am)$$

$$(a-b)m^3 + m(2a-2b) = mc$$

$$m((a-b)m^2 + 2a-2b-c) = 0$$

for common normal

$$m = 0 \text{ or}$$

$$m^2 = \frac{-(2a-2b-c)}{a-b}$$

$$m = \sqrt{-2 + \frac{c}{a-b}}$$

$$-2 + \frac{c}{a-b} > 0$$

$$\frac{c}{a-b} > 2$$

$$c > 2(a-b)$$

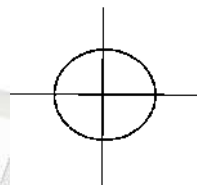
option (b)

4. (a) $X = \{(x,y): x^2 + 2y^2 = 3\}$ $x :$

$$y = \{(x,y): x > y, x, y \in \mathbb{Z}\}$$

$$x^2 + 2y^2 = 3$$

$$\frac{x^2}{3} + \frac{y^2}{3/2} = 1$$



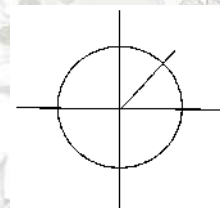
$$X = \{(1,1)(-1,1)(1,-1)(-1,-1)\}$$

$$y = \{x(2,-x)(x,y)\}$$

$$X \cap Y = \{(1,-1)\}$$

option (a)

5. (c)



$$x^2 + y^2 < 4$$

P probability of (r,s) $r > s$

$$P = (r,s)$$

first quadrant satisfying

$$P(E) = 1/2$$

6. (b) $f(x) = \frac{x}{e^{\pi x - 1}}$

$$f(0) = \frac{0}{e^{0-1}} = \frac{0}{e^{-1}} = \frac{0}{1/e} = 0$$

$$f(0) = 0$$

L.H.L.

$$\lim_{x \rightarrow 0^-} \left(\frac{x}{e^{\pi x - 1}} \right)$$

$$\lim_{h \rightarrow 0} \left(\frac{0-h}{e^{\pi(0-h)-1}} \right) = \lim_{h \rightarrow 0} \left(\frac{-h}{e^{\pi h - 1}} \right)$$

$$= \frac{0}{e^{-1}} = 0$$

R.H.L.

$$\lim_{x \rightarrow 0^+} \left(\frac{x}{e^{\pi x - 1}} \right) = \frac{0}{e^{-1}} = 0$$

$$f(0) = \text{R.H. L.} = \text{L.H.L.}$$

$$f(x) = \frac{x}{e^{\pi x - 1}}$$

$$f(x) = \frac{e^{\pi x - 1} \times \frac{d}{dx} x - x \frac{d}{dx} e^{\pi x - 1}}{(e^{\pi x - 1})^2}$$

$$f(x) = \frac{e^{\pi x - 1} x e^{\pi x - 1}}{(e^{\pi x - 1})^2}$$

$$\frac{e^{\pi x - 1} (1 - \pi x)}{(e^{\pi x - 1})^2}$$

$$f(x) = \frac{1 - \pi x}{e^{\pi x - 1}}$$

$$f(0) = \frac{1-0}{e^{-1}} = \frac{1}{1/e} = 0$$

$f(x)$ is continuous and differentiable function.



7. (c) $f(x) = x^{2/3} \cdot (6-x)^{1/3}$

$$f'(x) = x^{2/3} \cdot \frac{1}{3} \cdot (6-x)^{-2/3} + 2/3 (6-x)^{1/3} x^{-1/3}$$

$$f'(x) = x^{-1/3} \cdot (6-x)^{-2/3} \cdot \left(-\frac{x}{3} + \frac{2}{3}(6-x)\right)$$

$$f'(x) = \frac{1}{x^{1/3}(6-x)^{2/3}} \left(-\frac{x}{3} + 4 - \frac{2x}{3}\right)$$

$$f'(x) = -\frac{1}{x^{1/3}(6-x)^{2/3}} (x-4)$$

f has a point of inflection at $x = 0$

8. (a)

$$A = (2x+1)\hat{i} + (x^2-6y)\hat{j} + (xy^2+3z)\hat{k}$$

$$\nabla \cdot A = 0 \rightarrow \text{Solenoidal}$$

$$\nabla \cdot A = -ve \rightarrow \text{sink}$$

$$\nabla \cdot A = +ve \rightarrow \text{source}$$

$$\nabla \cdot A = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right) \cdot (2x+1)\hat{i} + (x^2-6y)\hat{j} + (xy^2+3z)\hat{k}$$

$$\frac{\partial}{\partial x}(2x+1) + \frac{\partial}{\partial y}(x^2-6y) + \frac{\partial}{\partial z}(xy^2+3z)$$

$$2 - 6 + 3 = -1 \quad (-ve \text{ sink})$$

9. (d)

$$A \cdot |\text{adj}(A)| \neq 0$$

if a is invertible matrix then $|A| \neq 0$

$$|\text{adj } A| = |A|^{n-1}$$

$$|\text{adj } A| \neq 0$$

$$|A| \neq 0$$

$$|AA^{-1}| = I$$

$$|A(\text{adj}(A))| \neq L$$

if $A = I$

then $|A| = 1$

$$|A(\text{adj}(A))| \neq L$$

so option d is incorrect.

10. (d)

$$\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x + 4^x}{4} \right)^{1/x}$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{1^x + 2^x + 3^x + 4^x - 1}{4} \right)^{1/x}$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{1^x + 2^x + 3^x + 4^x - 4}{4} \right)^{1/x}$$

$$\lim_{x \rightarrow 0} \left(1 + \frac{1^x + 2^x + 3^x + 4^x - 4}{4} \right)$$

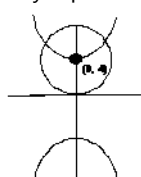
$$\lim_{x \rightarrow 0} \frac{1}{4} \left(\frac{1^x-1}{x} + \frac{2^x-1}{x} + \frac{3^x-1}{x} + \frac{4^x-1}{x} \right)$$

$$\Rightarrow e^{1/4(\log 1 \times 2 \times 3 \times 4)}$$

$$= (24)^{1/4} = (4!)^{1/4}$$

11. (c) $\frac{y^2}{9} - \frac{x^2}{16} = 1$ and $\frac{x^2}{4} + \frac{(y-4)^2}{16} = 1$

Only 2 point



12. (d) $\tan\left(\frac{\pi}{4} + \theta\right) \tan\left(\frac{3\pi}{4} + \theta\right)$

$$\tan\left(\frac{\pi}{4} + \theta\right) \tan\left(\frac{\pi}{2} + \frac{\pi}{4} + \theta\right)$$

$$\Rightarrow \tan\left(\frac{\pi}{4} + \theta\right) - \cot\left(\frac{\pi}{4} + \theta\right)$$

$$= -1$$

13. (c) $x + 2y + 2z = 5$

$$x + 2y + 3z = 6$$

$$x + 2y + 3z = \mu$$

$$A x = B$$

$$\begin{bmatrix} 1 & 2 & 2 & : & 5 \\ 1 & 2 & 3 & : & 6 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 2 & : & 5 \\ 0 & 0 & 1 & : & 1 \\ 0 & 0 & \sqrt{-2} & : & \mu - 5 \end{bmatrix}$$

For many solution, Rank (AB) order of matrix

$$\text{So } \lambda - 2 = 0, \lambda = 2, \mu - 5 = 0, \mu = 5$$

14. (c)

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{1 - \cos x} = \frac{0}{0}$$

L Hospital Rule

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2}{\theta + \sin x} = \frac{0}{0}$$

Again L hospital rule

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 0}{\cos x}$$

$$= 0.$$

15. (c)

$a^2x + ay + 1 = 0$ for real no. of a , is normal to the curve $xy = 1$

$$xy = L$$

$$x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

Normal

$$\frac{-dy}{dx} = \frac{x}{y} = \frac{x}{1/x} = x^2$$

$$y = ax - \frac{1}{a}$$

$$x^2 = -a$$

$$a < 0$$

16. (c) Total thrown = 8

$$\text{Odd no. of thrown} = (1, 3, 5, 7) = 4$$

$$P(E) \text{ of getting head in odd thrown} = \frac{1}{2}$$

17. (b)

$$\text{Total patient} = 3 (A, B, C)$$

$$P(E) \text{ of recovering a patient} = 0.6$$

$$P(\bar{E}) \text{ not recovering a patient} = 0.4$$

$$P(E) = P(A) \cdot P(B) \cdot P(\bar{C}) + P(A) \cdot P(\bar{B}) \cdot P(C)$$

$$+ P(\bar{A}) \cdot P(B) \cdot P(C)$$

$$\Rightarrow (0.6 \times 0.6 \times 0.4) + (0.6 \times 0.4 \times 0.6)$$

$$+ 0.9 \times 0.6 \times 0.6$$

$$= 3 \times 0.36 \times 0.4$$

$$= 0.432$$



18. (b) The value of $f(1)$ for $f\left(\frac{1-x}{1+x}\right) = x + 2$ is

let $x = 0$

$$f\left(\frac{1-0}{1+0}\right) = 0 + 2$$

$$f(1) = 2$$

19. (c) $x^2 - 2^y$

Solution of the equation

$$x^2 = 2^y \text{ are}$$

$$(1, 0) \text{ and } (2, 2)$$

The cardinality of the set C is 2 Ans.

20. (a) $f(x) = \cos[\pi^2 2] x + \cos[-\pi^2 2] [.] \rightarrow \text{GIF}$

$$f(x) = \cos[9.85] R + \cos[-9.85] x$$

$$f(x) = \cos 9x + \cos 10x$$

$$f\left(\frac{\pi}{2}\right) = \cos \frac{9}{2} \pi + \cos 5\pi$$

$$0 - 1$$

$$= -1$$

21. (d)

$\sin x = \sin y$ and $\cos x = \cos y$ then value of $x - y$ is

$$\cos(x-y) = \cos x \cdot \cos y + \sin x \cdot \sin y$$

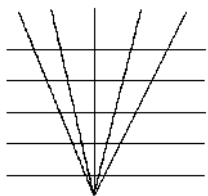
$$= \cos x \cdot \cos x + \sin x \cdot \sin x$$

$$= \cos^2 x + \sin^2 x$$

$$\cos(x-y) = 1$$

$$x - y = 2x\pi$$

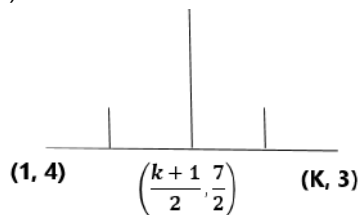
22. (c)



Total intersection $5 \times 5 + 1$

$$= 26$$

23. (a)



Eq. of perpendicular bisector

$$\left(y - \frac{7}{2}\right) = \left(\frac{k-1}{3-4}\right) \left(x - \frac{k+1}{2}\right)$$

$$y - \frac{7}{2} = (K-1) \left(x - \frac{K+1}{2}\right)$$

for y intercept $x = 0$

$$y - \frac{7}{2} = (K-1) \left(x - \frac{K+1}{2}\right)$$

$$\frac{2y-7}{2} = \frac{(1-K)(1+K)}{2}$$

$$2y - 7 = 1 - K^2$$

$$K^2 = 8 - 2 \times |4|$$

$$= 8 - 8 = 0 \text{ or } 8 + 8 = 16$$

$$K = 0 \text{ or } K = \pm 4$$

24. (a) $x(M \cup P \cup C) = 50$ $n(M) = 37$, $n(P) = 24$

$$n(C) = 43, n(M \cap P) = 19, n(M \cap C) = 29$$

$$n(P \cap C) = 20, n(A \cap B \cap C) = ?$$

$$n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(P \cap C)$$

$$n(C \cap P) + n(P \cap C \cap M)$$

$$50 = 34 + 24 + 43 - 19 - 20 + n(P \cap C \cap M)$$

$$P(P \cap C \cap M) = 14$$

25. (a) $AM = a$, $GM's = P$ and q

Let 2 no. 1, 8

$$AM a = 9/2$$

$$GM P, q = 2, 4$$

$$P^3 + q^3 = 8 + 64 = 72$$

$$P^3 + q^3 = 2pqa$$

26. (c)

Divided by 9, 10, 15 and 20 rem = 4, 5, 10

$$LCM = 9 \times 5 \times 4$$

$$= 180$$

$$\text{The no.} = 180n - 5, n = 2$$

$$\text{No} = 180 \times 2 - 5$$

$$= 355$$

27. (a) Mean, Median Mode = 1, 3^x , 9^x

$$\text{Mode} = 3 \text{ median} - 2 \text{ mean}$$

$$9^x = 3 \times 3^x - 2 \times 1$$

$$(3^x)^2 = 3 \times 3^x - 2$$

$$\text{Let } 3^n = x$$

$$x^2 - 3x + 2 = 0$$

$$(x+1)(x-2) = 0$$

$$x = 1, 9^x = 1$$

$$x = 2, 9^x = 4$$

$$\text{Mode} = 1, 4$$

28. (d)

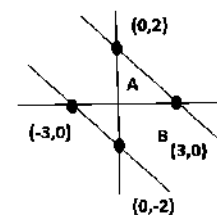
29. (a) The point $(1, 1/2)$ and $(3, -1/2)$ are

$$\text{Lines} = 2x + 3y = 6$$

$$= 2x + 3y = -6$$

$$A = (1, 1/2)$$

$$B = (3, -1/2)$$



Between the lines.

30. Foci $(4, 3)$ and $(1, 2, 5)$

$$\text{Distance between Cofii} = 2a e$$

$$SS' = 2ac = (12.4)^2 + (5.3)^2$$

$$\sqrt{64 + 4} = \sqrt{125}$$



$$2ae = 5\sqrt{5} \quad \sqrt{68}$$

$$ae = \sqrt{17}$$

$$OS' + OS' = 2a$$

$$\sqrt{16+9} + \sqrt{144+25} = 2a$$

$$2a = 5 + 13$$

$$A = 9$$

$$Ac = \sqrt{17}$$

$$e = \frac{\sqrt{17}}{a} = \frac{\sqrt{17}}{9}$$

31. (c)

Value of the series $\frac{2}{3!} + \frac{4}{5!} + \frac{6}{5!} + \dots$

$$\sum_{n=1}^{\infty} \frac{2n+1-1}{2n!+1}$$

$$\sum_{n=1}^{\infty} \frac{2n+1}{2n!+1} - \frac{-1}{2n!+1}$$

$$\sum_{n=1}^{\infty} \frac{1}{2n!+1} - \frac{-1}{2n!+1}$$

$$\Rightarrow 1 + 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} - \frac{1}{6!} + \dots$$

$$= e^{-1}.$$

32. (a) 9 bottle labelled 1,2,3....9

9 boxes labelled 1,2,39

5 bottle go their corresponding no = 9C_5

And other 4 bottle are not in their corresponding no = D4

$$= 4! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right]$$

$$= 4! \left[\frac{12-4+1}{24!} \right]$$

$$= 9$$

$$\text{Total ways} = 9 \times {}^9C_5$$

33. If married couple together total way = 7C_3

If married couple neither lengths = 7C_5

$$P(E) = \frac{{}^7C_3 + {}^7C_5}{{}^9C_5} = \frac{4}{9}$$

34. (c)

$$x = 1 + 2^{1/6} + 2^{2/6} + 2^{3/6} + \dots 2^{5/6}$$

$$a = 1, r = 2^{1/6}$$

$$S_5 = \frac{1(1-(2^{1/6})^6)}{1-2^{1/6}} = \frac{-1}{1-2^{1/6}} = \frac{1}{2^{2/6}-1}$$

$$\text{Value of } \left(1 + \frac{1}{x}\right)^{24} = 2^4 = 16$$

35. (a) $6x^2 - xy + \lambda y^2 = 0$

$$a = 6, b = \lambda, h = -1/2$$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a+b}$$

$$\text{for perpendicular line } a+b=0 \quad 6 + \lambda = 0$$

$$\lambda = -6$$

Of inclined at an angle of π

$$\tan \frac{\mu}{4} = \frac{2\sqrt{\frac{1}{4}-6\lambda}}{6+\lambda}$$

$$6 + \lambda = \frac{2\sqrt{1-24\lambda}}{2}$$

Squaring both side

$$(6+\lambda)^2 = (1-24\lambda)$$

$$\lambda^2 + 36 + 12\lambda = 1-24\lambda$$

$$\lambda^2 + 35 + 36\lambda$$

$$(\lambda + 35(\lambda + d)) = 0$$

$$\lambda = -35 \text{ or } \lambda = -1$$

$$36. |F| = 40N$$

$$|D| = 3M$$

$$\theta = 60^\circ$$

$$W = |F| |D| \cos \theta$$

$$= 40 \times 3 \times \frac{1}{2}$$

$$D = 60J \text{ ans.}$$

37. (c) Total no. divided by 2 and 3

$$\text{Both} = 6, 12, 18, \dots, 96$$

$$\text{Total} = 16$$

$$\text{Total natural no.} = 100$$

$$P(E) = \frac{{}^{16}C_3}{{}^{100}C_3}$$

$$\frac{16 \times 15 \times 14}{100 \times 99 \times 98}$$

$$\frac{4}{1155}$$

38. (d)

$$V_p = 4$$

$$|4| = \begin{vmatrix} m & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 2 & -1 \end{vmatrix}$$

$$|4| = m(-1)(-1) - (-2) + 1(3)$$

$$|4| = -m + 5$$

$$+4 = -m + 5$$

$$M = 1, 9$$

$$39. (c) f(x) = \begin{cases} x^2 \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} \Rightarrow 0$$

Continuous

$$\lim_{x \rightarrow 0} \frac{1(x) - (0)}{n-0}$$

$$= \frac{x^2 \sin(\frac{1}{x}) - 0}{x}$$

$$\Rightarrow 0 \text{ differentiable}$$

$$f'(x) = x^2 \cdot \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right) + \sin \frac{1}{x} \cdot 2x$$

$$f'(x) = -\cos \frac{1}{x} + 2x \sin \frac{1}{x}$$

$$\lim_{x \rightarrow 0} f'(x)$$

$$X = 0$$

$$-\cos \frac{1}{x} + 2x \sin \left(\frac{1}{x}\right)$$

Oscillating + o x osall

= oscillating

$f'(x)$ not continuous at $x = 0$

$$40. (b) \int_a^3 e^{x^2} dx = \int_a^5 e^{x^2} dx + \int_5^3 e^{x^2} dx$$

$$\int_a^b f(x) = \int_a^c f(x) + \int_c^b f(x)$$

Option (b) is correct.

41. (d) $m_1 = 2 \rightarrow A$

$$m_2 = 4 \rightarrow B$$

Median of combined series

$$M \geq \min(M_1, M_2)$$

$$M \geq 2$$



42. (b)

$$\sum_{r=1}^n \frac{1}{2^n} \frac{n P_r}{r!}$$

Let $n = 2, 1$

$$\sum_{r=1}^n \frac{1}{2^n} \frac{2 P_r}{r!}$$

$$\frac{1}{4} \left[\frac{2 P_1}{1!} + \frac{2 P_2}{2!} \right]$$

$$\frac{1}{4} \left[\frac{2!}{1!} + 1 \right]$$

$$\frac{3}{4}$$

43. (d) $f: (1, 2, 3) \rightarrow \{a, b, c, d, e\}$

Total one-one

$$= 5 \times 4 \times 3 = 60$$

44. (b) $C = \left\{ x \mid \sin 4x = \frac{1}{2}, x \in (-9\pi, 3\pi) \right\}$

$$x \in (-9\pi, 3\pi)$$

$$4\pi \in (-36\pi, 12\pi)$$

$$0 - 2\pi \quad \sin 4x = \frac{1}{2} \quad \text{for 2 value of } x$$

$$-36, 12\pi \quad \sin 4x = \frac{1}{2} \quad \text{for 48 value}$$

$$48$$

45. (a)

$$|d| = 25 \text{ m}$$

$$|F| = 180$$

$$\theta = 45^\circ$$

$$W = |F| |d| \cos \theta$$

$$= 25 \times 180 \times \frac{1}{\sqrt{2}}$$

$$\frac{45000}{1.4}$$

$$= 3.18 \times 10^3 \text{ J}$$

46. (b)

$$cf(x) + d f(1/x) = |\log |x|| + 3$$

$$c^2 f(x) + cd f(1/x) = c |\log |x|| + 3c$$

$$d^2 f(x) + cd f(1/x) = d |\log |x|| + 3d$$

Equation (i) and (ii) subtract

$$(c^2 - d^2) f(x) = (c - d) |\log |x|| + 3(c - d)$$

$$(c^2 - d^2) f(x) = (c - d)(|\log |x|| + 3)$$

$$f(x) = \frac{c - d}{c^2 - d^2} (|\log |x|| + 3)$$

$$\frac{c - d}{c^2 - d^2} \int_1^e |\log |x|| + 3$$

$$\frac{c - d}{c^2 - d^2} \int_1^e \log x + 3 = \frac{c - d}{c^2 - d^2} [x \log x - x + 3x]_1^e$$

$$= \frac{(c - d)(3c - 2)}{c^2 - d^2}$$

47. (b) $3x^2 + 10xy + 11y^2 + 148$

$$+ 12y + 5 = 0 \dots\dots$$

a....

$$a = 3, h = 5, b = 11, g =$$

$$f = 6, C = 5$$

$$h^2 - ab$$

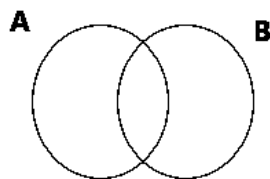
$$25 - 33$$

$$-8 < 0$$

$$h^2 - ab < 0, \quad \text{ellipse}$$

48. (d)

$$P((A \cap B^c) \cup (A^c \cap B))$$



$$\Rightarrow P(A \cup B) - P(A \cap B)$$

$$= P(A) + P(B) - P(A \cap B)$$

$$- P(A \cap B)$$

$$= P(A) + P(B) - 2P(A \cap B)$$

$$49. (b) \left. \begin{array}{l} \lambda^2 i + j + k \\ i + \lambda^2 + K \\ i + n + \lambda^2 K \end{array} \right\} \text{coplanar}$$

$$\begin{vmatrix} \lambda^2 & 1 & 1 \\ 1 & \lambda^2 & 1 \\ 1 & 1 & \lambda^2 \end{vmatrix} = 0$$

$$\lambda^2 (\lambda^4 - 1) - 1 (\lambda^2 - 1) + 1 (1 - \lambda^2) = 0$$

$$\lambda^6 - \lambda^2 - \lambda^2 + 1 + 1 - \lambda^2 = 0$$

$$\lambda^6 - 3\lambda^2 + 2 = 0$$

$$\lambda^6 - \lambda^2 - 2\lambda^2 + 2$$

$$\lambda^2 (\lambda^4 - 1) - 2(\lambda^2 - 1)$$

$$\lambda^2 (\lambda^2 - 1) (\lambda^2 + 1) - 2(\lambda^2 - 1)$$

$$(\lambda^2 - 1) (\lambda^4 + \lambda^2 - 2) = 0$$

$$\lambda = \pm$$

2 value of λ

50. (b) $P(A) = 40\% = 4/10$

$$P(B) = 50\% = 5/10$$

$$P(A) \cdot P(\bar{B}) + P(B) \cdot P(\bar{A})$$

$$\frac{4}{10} \times \frac{5}{10} + \frac{5}{10} \times \frac{6}{10}$$

$$\frac{50}{100} = \frac{1}{2}$$

REASONING

51. (a)

9 Red

2 Uce = Reduce

5 Exp

7 And = Expand

52. (c)

53. (c)

Years of experience show the number like 1 year

Salary show the number like 10000

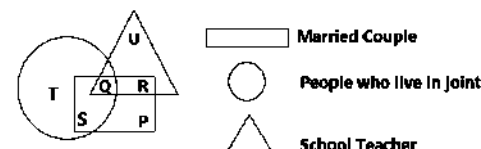
Age show the number like 10 years

But Highest Education not show the number.

54. (b)

His cousin

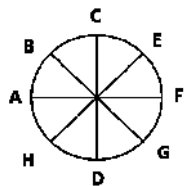
55. (c)





The married teacher who do not live in joint family are represented – R.

56. (d) F



57. (a)

$$\begin{aligned} 2^3 &= 8 \\ 2^2 &= 4 \\ 3^3 &= 27 \\ 3^2 &= 9 \end{aligned}$$

So (a) is the correct option.

58. (c)

If you have no data you are not make the graph.

You have no milk are are not make the butter.

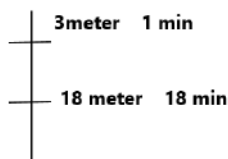
So option (c) is correct.

59. (c)

Pole length 21 meter

3 meter 2 meter 5 meter 4 meter

1 min 2 min 3 min 4 min



60. (d)

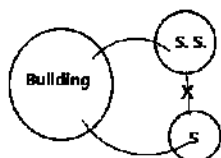
Data	1
Software	2
Code	3
Analysis	4
Report	5

61. (c)

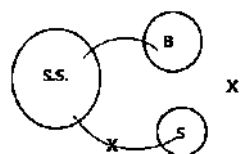
Option (a)

A C F

X

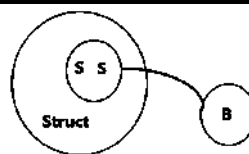


Option (b) A C E



Option (c)

BDF



Ans. BDF

62. (b)

HEM from center - 3 + 8

JGP from center -3 + 9

OLT from center -3 + 8

NKS from center -3 + 8

So (b) option is odd.

63. (d)

Car A start point x and travels at a constant speed

Speed = 60 km/h time = 1.5 hr

Car B starts point x

Speed = 80 km/h time = 1.5 hr

Car A

$$S = \frac{D}{T} = 60 \Rightarrow \frac{D}{1.5} \Rightarrow D = 90.0 \text{ km}$$

Car B

$$80 = \frac{D}{1.5} \Rightarrow D_2 = 120.0 \text{ km}$$

Difference in distance covered by Car A compound to car

$$120 - 90$$

$$= 30 \text{ km}$$

Option (d)

64. (a)

100 → MP

↓ -20%

80 → S.P.

100 → C.P.

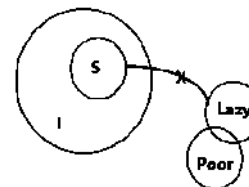
111.11% 80

$$1\% = \frac{80}{111.11} \times 100$$

→ ~72

$$\frac{28}{72} \times 100 = 38.88\%$$

65. (d)



i. No student is lazy = True

ii. same poor people are not intelligent = True

iii. All poor people are lazy = Doubt

Only (i) and (ii) follows.

Option (d)

66. (d)

Arjun is 25 years younger to his mother

His brother is 35 years younger to his mother

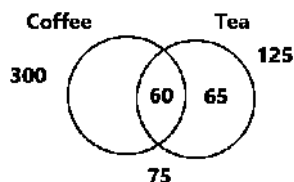
His brother Birth year 1964

Arjun's Birth year is 1964 - 10 = 1954



67. (c)
- | | | | |
|---|---|---|-------|
| A | ✓ | x | truth |
| B | x | x | |
| C | ✓ | ✓ | |

68. (c)
- Total = 500
- $h(c) = 60\% = 300$
- $n(T) = 25\% = 125$
- 15% have no preference
- $20\% = (CAT) \Rightarrow \frac{300 \times 20}{100} = 60$



65 Ans.

69. (a)
- HEART 2018010508
- H alphabet no. is 8 in last
- LUNGS 1907142112
- L alphabet no. is 12 in last
- BRAIN
- B alphabet no. is 2 in last

70. (a)
- The mean of positive consecutive integer from 2 to n is
- $2 + 3 + \dots + n \rightarrow AP$
- (n-1) terms
- $$S_n = \frac{(n-1)}{2} [2 \times 2 + (n-1-1)1]$$
- $$= \frac{(n-1)}{2} (4 + n - 2)$$
- $$= \frac{(n-1)}{2} (n+2)$$
- Mean = $\frac{(n-1)(n+2)}{2(n-1)} = \frac{n+2}{2}$
- Option (a)

71. (d)
72. (d)
- BATMINTON
- In simple image BATMINTON
- In mirror image NOTNIMTAB
- So the option (d) is correct.

73. (b)
- Which is the value of $x^2 + y^2 = ?$
- Statement (I) $xy = 5$
- Statement (II) $x + y = 10$
- $(x + y)^2 = x^2 + y^2 + 2xy$
- $100 = x^2 + y^2 + 10$
- $x^2 + y^2 = 90$
- Option (b)

74. (c)

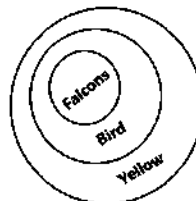
Aditi	Bharat	Chandan	Deepak
Burger	Salad	Pasta	Pizza
Mango Juice	X		X
	Lemon		Orange juice

Who ordered the Burger and what drink did they order

Aditi, Orange Juice

Ans. Option (c)

75. (b)



76. (b) In half year Exam
- Passed student = 60%
- In Annual exam passed student = 60% of 70%
- = 42%
- Remaining student 80% passed in annual exam
- = 40% of 80%
- = 32%
- Passed student in annual exam = 42% + 32%
- = 74%

Option (b)

77. (d)
- Odometer $\rightarrow$ Mileage
- Compass $\rightarrow$ Direction

78. (b)
- Option (b) Ans.
- Compline

79. (c)
- $100 \times 50 = 5000$
- $5000 \rightarrow 100 = 5100$
- $100 \times 55 = 5500 \rightarrow -2\% = 110$
- $5500 - 110 = 5390$
- $5390 - 5100 = \frac{290}{5100} \times 100$
- $\frac{290}{51} = 5.69\%$

80. (b)
- | | |
|----|----|
| R | S |
| 15 | 20 |

60 days

They delhi 5 days back

How many days from today they will met

$60 - 5 = 55$ days

Option (b)

81. (b)
- | | | |
|-----------------------------|---|----|
| 2 | 6 | 2x |
| 80 th March 1992 | | |
| 0 | 4 | 0 |
| = Monday | | |

Option (b)

82. (b)

83. (d) $(2 \text{ ----- } 3) 180^\circ$
- $\downarrow$
- 8 (40 min)



$$\text{Min} + \frac{\text{min}}{11}$$

$$40 + \frac{40}{11}$$

$$2 : 43 \frac{7}{11}$$

Option (d)

84. (d)

Swift	Creta	Nexon	D' doc
B+G	AF-D	E* C-to c	C- doc x A+ F
Teacher	Teacher	Teacher	X
	English		Eng.

What is profession

= Engineer

Option (d)

85. (c) Nexon

86. (b) GEA

87. (a)

$$X + 32 \quad D + 14$$

$$X + 34 \quad C + 6$$

$$X + 16 \quad B - 2$$

$$X + 8 \quad E - 10$$

$$Z \leftarrow A - 18$$

$$A + E + B + C + D = E$$

$$X + x + 16 + x + 24 + x + 32 = 0$$

$$4x = -72$$

$$X = -18$$

88. (d)

$$A + B$$

$$x + x + 16$$

$$2x + 16$$

$$-36 + 16$$

$$= -20$$

89. (a)

$$\text{Greatest number } x + 32$$

$$-18 + 32$$

$$= 14$$

Option (a)

90. (b)

$$A + B + C + D + E$$

$$x + x + 8 + x + 16 + x + 24 + x + 32$$

$$5x + 80$$

$$5x(-18) + 80$$

$$-90 + 80$$

$$= -10$$

Option (b)

COMPUTER

91. (d)

$$\begin{array}{r} \underline{101011} \\ 101) 11010111(\\ \underline{101} \\ 00110 \\ \underline{101} \\ 00111 \\ \underline{0101} \\ 101 \\ \underline{101} \\ 000 \end{array}$$

92. (d)

Any given truth table can be represented by

Option (d) All of the above

93. (a) DMA

94. (c)

CPU and RAM

95. (d)

32 bit

$$\text{Total memory} = 2^{32}$$

$$\text{Page size} = 4 \text{ KB} = 2^{12} \text{ bit}$$

$$\text{Total page} = \frac{2^{32}}{2^{12}} = 2^{20}$$

$$1 \text{ Entry} = 128 \text{ page} \Rightarrow 1 \text{ set (for tag)} = \frac{2^7}{2^2} = 2^5 \text{ page}$$

$$\text{Total Entry} = \frac{2^{20}}{2^5} = 2^{15} \text{ page}$$

$$\text{Tag bit} = 15 \text{ bit Ans.}$$

Option (d) is correct answer.

96. (b)

The expression $P + \overline{0}R$ is reduced from option (b)

$$(P + \overline{0}) (P + R)$$

$$P + PR + P\overline{0} + \overline{0}R$$

$$P(1 + R + \overline{0}) + \overline{0}R$$

$$P(1 + \overline{0}R) = P + \overline{0}R$$

97. (a)

10 temporarily store frequently accessed data and instruction for faster access by CPU.

98. (d)

$$a = 10$$

$$b = 20$$

$$c = 30$$

$$d = a + c$$

$$e = b + d$$

$$f = c + e$$

$$b = c + e$$

$$e = b + f$$

$$d = 5 + e$$

$$\text{return } d + f$$

d replace a

f replace b

$$a = a + c$$

$$a = b + a$$

$$b = c + a \rightarrow 3 \text{ variable us.}$$

$$d = b + b$$

Option (d) 3

99. (a)

Which of the following is the smallest unit of data in a computer.

100. (a)

Consider the following 4-bit binary number represented in the 2's complement from ; 1101 and 0100

$$A = 1101 \Rightarrow 0011$$

$$= -3$$

$$B = 0100 \Rightarrow 4$$

$$A + B = 4 - 3 = 1$$

$$1 = 0.001$$

No over flow.

Option (a) 0001 and no overflow.



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101. (b)
 RAM = 64 K Byte
 Exist RAM = 4K x 8 bit

$$\text{Total} = \frac{64 \text{ K Byte}}{4 \text{ K} \times \text{Byte}}$$
 = 16 chips
 Option (b)

102. (c)
 A = 11111111 2's com. 00000001 = -1
 B = 11111111 2's com. 00000001 = -1
 -2 = 00000010
 2's = 11111110

103. (c)

x	y	z	f
0	0	0	0
0	0	1	0
$\sqrt{0}$	1	0	1
0	1	1	0
$\sqrt{1}$	0	1	1
1	1	0	0
1	1	1	1

$$F = x y z' + xy'z + xyz$$

Option (c)

104. (c)
 Solution
 00, 01, 0A, 10, 11, 1A, A0, A1, AA, 100, 101, 10A
 10A = 12th Position

105. (b) Register file

106. (c) - $1022 \leq E \leq 1023$
 107. (c) memory data Register.
 108. (b) 8

109. (c)
 Amount of RAM

110. (b)
 2's 11001 = 00111 = -7
 2's 1001 = 0111 = -7
 2's 111001 = 000111 = -7
 Option (b) -7, -7 and -7 respectively.

ENGLISH

111 (a)	112 (a)	113.(c)	114 (b)	115.(a)
116.(a)	117 (d)	118.(c)	119. (b)	120. (d)

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